

Practical File Submission

Course Name: JavaScript Lab

Course Code: T-3013 / 0705210508

Submitted by

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24070521161

Batch: 2024-28

Semester - V

Section: C

Submitted to

Dr. Shreyas Rajendra Hole

Practical no 1

Experiment Title:

Demonstrate inline, internal, and external JavaScript; use console methods; create a webpage showing user info and a welcome message.

Software/Tools Required:

Vs code , chrome browser

Demo html file:

```
<!DOCTYPE html>
<html>
  <head>
    <title> internal css</title>

    <!-- External CSS -->
    <link rel="stylesheet" href="style.css">

    <style>
      h1{
        color:red;
        font-size:24px;
        text-align :center;
      }

      p{
        color:blue;
        font-size:20px;
        text-align:center;
      }
      .btn{
        display:flex;
        justify-content:center;
        align-items:center;
      }
      button{
        background-color:green;
        color:white;
        padding:10px;
        position :absolute;
        margin: left;
        border:none;
        border-radius:5px;
      }
    </style>
  </head>

  <body>

    <!-- Inline CSS -->
    <h1 style="background-color:yellow;"> Welcome to internal css</h1>

    <p>Ayush katekhaye</p>

    <p>24070521161</p>

    <div class="btn">
```

```
<button onclick="showMessage()">click me</button>
</div>

<br><br><br>

<div class="btn">
  <button onclick="displayInfo()">External JavaScript</button>
</div>

<!-- Text using External CSS -->
<h2 class="external">This text is styled using External CSS</h2>

<!-- Internal JavaScript -->
<script>

  function showMessage()
  {
    alert("Welcome to Internal JavaScript!");

    console.log("This is console.log()");

    console.info("This is console.info()");

    console.warn("This is console.warn()");

    console.error("This is console.error()");

    console.table([
      {
        Name:"Ayush Katekhaye",
        RollNo:"24070521161"
      }
    ]);
  }

</script>

<!-- External JavaScript -->
<script src="script.js"></script>

</body>
</html>
```

style.css

```
.external{
  color:purple;
  text-align:center;
  font-size:24px;
  font-weight:bold;
  margin-top:80px;
}
```

Script.js

```
function displayInfo()
{
  alert("Welcome to External JavaScript!");

  console.log("External JavaScript Executed");

  let student = {
```



```
Name: "Ayush Katekhaye",  
RollNo: "24070521161",  
Branch: "Computer Science Engineering"  
};  
  
console.table(student);  
}
```



Welcome to internal css

Ayush katekhaye

24070521161

click me

External JavaScript

This text is styled using External CSS

Case study 1

Case Study Title:- Create a web page for college department and link the student information page with it and write a js program to display "Welcome to SIT Nagpur" in an alert box

Case Study Code:

department.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>SIT Nagpur - Department</title>
  <style>
    :root {
      --primary: #1f4d7a;
      --accent: #f7b733;
      --light: #fafafa;
      --dark: #102c42;
      --text: #273444;
      --surface: rgba(255,255,255,0.92);
    }

    * {
      box-sizing: border-box;
    }

    html {
      scroll-behavior: smooth;
    }

    html,
    body {
      margin: 0;
      padding: 0;
      min-height: 100%;
      font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
      color: var(--text);
      background: linear-gradient(180deg, rgba(15, 23, 42, 0.78), rgba(15, 23, 42, 0.78)),
        url("sit_bg.jpg") center/cover no-repeat fixed;
      background-blend-mode: overlay;
    }

    body {
      display: flex;
      flex-direction: column;
    }
```



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```
.page-wrapper {
  width: min(1100px, 92%);
  margin: 0 auto;
  padding: 24px 0 36px;
  opacity: 0;
  animation: pageFadeIn 0.9s ease 0.2s forwards;
}

.site-header {
  position: sticky;
  top: 0;
  z-index: 20;
  display: flex;
  align-items: center;
  justify-content: space-between;
  gap: 16px;
  padding: 18px 0 16px;
  flex-wrap: wrap;
  background: rgba(9, 18, 34, 0.86);
  backdrop-filter: blur(14px);
  border-bottom: 1px solid rgba(255, 255, 255, 0.08);
  box-shadow: 0 18px 40px rgba(0, 0, 0, 0.18);
  animation: fadeInDown 0.8s ease 0.1s both;
}

.site-brand {
  display: flex;
  align-items: center;
  gap: 16px;
}

.site-nav {
  display: flex;
  align-items: center;
  gap: 12px;
  flex-wrap: wrap;
}

.nav-link {
  color: rgba(255, 255, 255, 0.92);
  text-decoration: none;
  padding: 10px 16px;
  border-radius: 999px;
  transition: background 0.2s ease, color 0.2s ease;
  font-weight: 600;
  font-size: 0.95rem;
}

.nav-link:hover {
  background: rgba(255, 255, 255, 0.14);
}
```



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```
.nav-button {
  background: rgba(255, 255, 255, 0.16);
  border: 1px solid rgba(255, 255, 255, 0.22);
}

.site-brand img {
  width: 120px;
  max-height: 120px;
  object-fit: contain;
  border-radius: 22px;
  box-shadow: 0 18px 38px rgba(0, 0, 0, 0.22);
  background: #ffffff;
  padding: 12px;
}

.site-brand h1 {
  margin: 0;
  font-size: 1.25rem;
  color: #fbfbfb;
  line-height: 1.2;
}

.site-brand p {
  margin: 6px 0 0;
  color: #d7e0f2;
  font-size: 0.98rem;
}

.hero {
  background: var(--surface);
  border-radius: 28px;
  padding: 40px;
  box-shadow: 0 24px 80px rgba(4, 20, 44, 0.18);
  backdrop-filter: blur(14px);
  animation: fadeInUp 0.9s ease 0.25s both;
}

.hero-visual {
  margin: 32px auto 0;
  min-height: 260px;
  border-radius: 28px;
  background: linear-gradient(180deg, rgba(31, 77, 122, 0.32), rgba(31, 77, 122, 0.1)),
url("sit_bg.jpg") center/cover no-repeat;
  border: 1px solid rgba(31, 77, 122, 0.12);
  box-shadow: 0 18px 42px rgba(15, 23, 42, 0.14);
  position: relative;
  overflow: hidden;
  animation: float 8s ease-in-out infinite, fadeIn 1s ease 0.35s both;
}
```



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```
.hero-visual::before {  
  content: "";  
  position: absolute;  
  inset: 0;  
  background: linear-gradient(180deg, rgba(0, 0, 0, 0.18), rgba(0, 0, 0, 0.12));  
}
```

```
.hero-visual__content {  
  position: absolute;  
  bottom: 20px;  
  left: 24px;  
  z-index: 1;  
  max-width: 360px;  
  color: #fff;  
  font-weight: 700;  
  font-size: 1.05rem;  
  line-height: 1.35;  
  background: rgba(0, 0, 0, 0.32);  
  padding: 14px 18px;  
  border-radius: 16px;  
  border: 1px solid rgba(255, 255, 255, 0.16);  
}
```

```
.hero h2 {  
  margin: 0;  
  font-size: clamp(2rem, 4vw, 3rem);  
  color: var(--dark);  
  line-height: 1.05;  
}
```

```
.hero p {  
  margin: 24px auto 0;  
  max-width: 760px;  
  line-height: 1.8;  
  font-size: 1.05rem;  
  color: #4b6377;  
}
```

```
.cta {  
  margin-top: 32px;  
  display: flex;  
  flex-wrap: wrap;  
  justify-content: center;  
  gap: 16px;  
}
```

```
.button,  
.button-secondary {  
  display: inline-flex;  
  align-items: center;  
  justify-content: center;
```



```
padding: 14px 26px;  
border-radius: 999px;  
text-decoration: none;  
font-weight: 700;  
transition: transform 0.2s ease, box-shadow 0.2s ease, background 0.2s ease;  
}
```

```
.button {  
  background: linear-gradient(135deg, var(--primary), #32678e);  
  color: #fff;  
  border: 1px solid rgba(255, 255, 255, 0.15);  
}
```

```
.button-secondary {  
  background: #ffffff;  
  color: var(--primary);  
  border: 1px solid rgba(31, 77, 122, 0.16);  
}
```

```
.button:hover,  
.button-secondary:hover {  
  transform: translateY(-2px);  
  box-shadow: 0 18px 36px rgba(31, 77, 122, 0.18);  
}
```

```
.toast {  
  position: fixed;  
  right: 24px;  
  bottom: 24px;  
  max-width: 320px;  
  padding: 14px 18px;  
  border-radius: 14px;  
  background: rgba(15, 23, 42, 0.94);  
  color: #fff;  
  font-size: 0.95rem;  
  box-shadow: 0 18px 40px rgba(0, 0, 0, 0.24);  
  opacity: 0;  
  transform: translateY(12px);  
  pointer-events: none;  
  transition: opacity 0.25s ease, transform 0.25s ease;  
  z-index: 9999;  
}
```

```
.toast.show {  
  opacity: 1;  
  transform: translateY(0);  
}
```

```
.hero-grid {  
  display: grid;  
  grid-template-columns: repeat(auto-fit, minmax(220px, 1fr));  
}
```



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```
gap: 18px;
margin-top: 38px;
}

.feature-card {
  background: #ffffff;
  border-radius: 22px;
  padding: 24px;
  min-height: 190px;
  display: flex;
  flex-direction: column;
  justify-content: center;
  border: 1px solid rgba(15, 23, 42, 0.08);
  box-shadow: 0 14px 32px rgba(15, 23, 42, 0.08);
  opacity: 0;
  transform: translateY(22px);
  animation: fadeInUp 0.8s ease both;
}

.feature-card:nth-child(1) { animation-delay: 0.45s; }
.feature-card:nth-child(2) { animation-delay: 0.60s; }
.feature-card:nth-child(3) { animation-delay: 0.75s; }

.feature-card h3 {
  margin: 0 0 10px;
  color: var(--primary);
  font-size: 1.1rem;
}

.feature-card p {
  margin: 0;
  color: #5d7388;
  line-height: 1.75;
}

@keyframes pageFadeIn {
  from { opacity: 0; transform: translateY(10px); }
  to { opacity: 1; transform: translateY(0); }
}

@keyframes fadeInDown {
  from { opacity: 0; transform: translateY(-18px); }
  to { opacity: 1; transform: translateY(0); }
}

@keyframes fadeInUp {
  from { opacity: 0; transform: translateY(24px); }
  to { opacity: 1; transform: translateY(0); }
}

@keyframes fadeIn {
```

```

    from { opacity: 0; }
    to { opacity: 1; }
  }

  @keyframes float {
    0%, 100% { transform: translateY(0); }
    50% { transform: translateY(-8px); }
  }

  footer {
    margin-top: auto;
    color: #d7e0f2;
    text-align: center;
    padding: 18px 0 10px;
    font-size: 0.98rem;
  }

  @media (max-width: 720px) {
    .site-header {
      flex-direction: column;
      align-items: flex-start;
    }

    .cta {
      justify-content: stretch;
    }

    .button,
    .button-secondary {
      width: 100%;
    }
  }
</style>
</head>
<body id="top" class="department-page">
  <div class="page-wrapper">
    <header class="site-header">
      <div class="site-brand">
        
        <div>
          <h1>Symbiosis Institute of Technology</h1>
          <p>Computer Science and Engineering Department</p>
        </div>
      </div>
      <nav class="site-nav">
        <a href="#top" class="nav-link clickable" data-toast="Returning Home...">Home</a>
        <a href="#highlights" class="nav-link clickable" data-toast="Jumping to
Highlights...">Highlights</a>
        <a href="student.html" class="nav-link nav-button clickable" data-toast="Opening Student
Info...">Student Info</a>
      </nav>

```

</header>

<section class="hero">

<h2>Build the future with AI, Cloud, Data Science and Cyber Security.</h2>

<p>The Department of Computer Science and Engineering at SIT Nagpur empowers students with hands-on learning, modern labs and industry-aligned projects for careers in Programming, Artificial Intelligence, Data Science, Cyber Security, Cloud Computing and Software Engineering.</p>

<div class="cta">

View Student Information

Explore Highlights

</div>

<div class="hero-visual" aria-label="Campus background image">

<div class="hero-visual__content">Experience our modern labs and campus environment with a focus on tech innovation.</div>

</div>

<div id="highlights" class="hero-grid">

<article class="feature-card">

<h3>Practical Learning</h3>

<p>Project-based labs and campus events give students real experience solving industry problems.</p>

</article>

<article class="feature-card">

<h3>Modern Technology</h3>

<p>Students work with AI, cloud platforms, cybersecurity tooling and full-stack software systems.</p>

</article>

<article class="feature-card">

<h3>Student Success</h3>

<p>Strong placement support, mentoring and a culture of innovation help learners achieve their goals.</p>

</article>

</div>

</section>

</div>

<footer>© 2026 Symbiosis Institute of Technology, Nagpur</footer>

<div id="toast" class="toast"></div>

<script src="script.js"></script>

</body>

</html>



॥ वसुधैव कुटुम्बकम् ॥

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student.html

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<meta name="viewport"  
content="width=device-width, initial-  
scale=1.0">
```

```
<title>SIT Nagpur - Student Information</title>
```

```
<style>
```

```
:root {
```

```
--blue: #115288;
```

```
--surface: #ffffff;
```

```
--bg: #edf5ff;
```

```
--text: #172d46;
```

```
--muted: #5a7188;
```

```
}
```

```
* {
```

```
box-sizing: border-box;
```



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```
}
```

```
body {
```

```
    margin: 0;
```

```
    min-height: 100vh;
```

```
    font-family: 'Segoe UI', Tahoma, Geneva,  
    Verdana, sans-serif;
```

```
    background: linear-gradient(135deg, #e5f2ff,  
    #f8fbff);
```

```
    color: var(--text);
```

```
}
```

```
.page-wrapper {
```

```
    width: min(980px, 92%);
```

```
    margin: 0 auto;
```

```
    padding: 28px 0 40px;
```

```
}
```

```
header {
```

```
    display: flex;
```

```
align-items: center;

justify-content: space-between;

gap: 16px;

flex-wrap: wrap;

margin-bottom: 28px;

}
```

```
.student-logo {

  width: 86px;

  height: auto;

  object-fit: contain;

  border-radius: 18px;

  background: #ffffff;

  padding: 8px;

  box-shadow: 0 14px 30px rgba(17, 82, 136,
0.16);

}
```

```
.student-heading {

  display: flex;
```



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align-items: center;

gap: 16px;

flex-wrap: wrap;

}

header h1 {

margin: 0;

font-size: clamp(2rem, 3.5vw, 2.4rem);

letter-spacing: 0.02em;

}

header p {

margin: 8px 0 0;

color: var(--muted);

max-width: 560px;

}

.card {

background: var(--surface);


```
border-radius: 24px;  
  
padding: 32px;  
  
box-shadow: 0 24px 60px rgba(17, 82, 136,  
0.12);  
  
border: 1px solid rgba(17, 82, 136, 0.06);  
  
}
```

```
.details {  
  
width: 100%;  
  
border-collapse: collapse;  
  
margin: 0;  
  
color: var(--text);  
  
}
```

```
.details th,  
.details td {  
  
padding: 16px 14px;  
  
text-align: left;  
  
vertical-align: top;  
  
}
```



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```
.details th {  
    width: 180px;  
    font-weight: 700;  
    color: var(--blue);  
    border-bottom: 1px solid #e5ecf5;  
}
```

```
.details td {  
    border-bottom: 1px solid #eef4fb;  
    color: #42586d;  
}
```

```
.details tr:last-child th,  
.details tr:last-child td {  
    border-bottom: none;  
}
```

```
.profile-card {
```

```
display: flex;

align-items: center;

gap: 18px;

background: #f8fbff;

border-radius: 24px;

padding: 24px;

margin-bottom: 26px;

border: 1px solid rgba(17, 82, 136, 0.12);

box-shadow: 0 18px 40px rgba(17, 82, 136, 0.08);

}
```

```
.male-avatar {

  position: relative;

  width: 150px;

  height: 240px;

  display: flex;

  align-items: flex-start;

  justify-content: center;

  padding-top: 10px;
```



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```
}
```

```
.male-head {
```

```
    position: relative;
```

```
    width: 130px;
```

```
    height: 130px;
```

```
    border-radius: 50%;
```

```
    background: #f7d4b5;
```

```
    border: 5px solid rgba(17, 82, 136, 0.16);
```

```
    box-shadow: inset 0 -8px 0 rgba(0, 0, 0, 0.06), 0 18px 40px rgba(17, 82, 136, 0.08);
```

```
    overflow: hidden;
```

```
}
```

```
.male-head::before {
```

```
    content: "";
```

```
    position: absolute;
```

```
    top: 0;
```

```
    left: 0;
```

```
    right: 0;
```

```
height: 60px;  
  
background: #3a2f24;  
  
border-bottom-left-radius: 60% 40%;  
border-bottom-right-radius: 60% 40%;  
  
}
```

```
.male-head::after {  
  
  content: "";  
  
  position: absolute;  
  
  bottom: 0;  
  
  left: 0;  
  
  right: 0;  
  
  height: 24px;  
  
  background: linear-gradient(180deg,  
    rgba(0,0,0,0.08), transparent);  
  
}
```

```
.male-eye {  
  
  position: absolute;  
  
  width: 18px;
```



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```
height: 18px;

background: #fff;

border-radius: 50%;

top: 56px;

box-shadow: inset 0 0 0 2px rgba(17, 82, 136,
0.18);

}
```

```
.male-eye::after {

content: "";

position: absolute;

width: 8px;

height: 8px;

background: #2e1f15;

border-radius: 50%;

top: 4px;

left: 4px;

}
```

```
.male-eye.left {
```

left: 32px;

}

.male-eye.right {

right: 32px;

}

.male-nose {

position: absolute;

top: 84px;

left: 50%;

width: 20px;

height: 24px;

background: #d99d73;

border-radius: 50% 50% 40% 40%;

transform: translateX(-50%);

box-shadow: inset 0 2px 0 rgba(0,0,0,0.06);

}



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```
.male-mouth {  
  
    position: absolute;  
  
    bottom: 22px;  
  
    left: 50%;  
  
    width: 40px;  
  
    height: 14px;  
  
    border-radius: 0 0 40px 40px;  
  
    border-bottom: 4px solid rgba(50, 24, 10,  
0.25);  
  
    transform: translateX(-50%);  
  
}
```

```
.male-brow {  
  
    position: absolute;  
  
    width: 34px;  
  
    height: 10px;  
  
    background: #2e1f15;  
  
    border-radius: 12px;  
  
    top: 44px;  
  
}
```




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```
.male-brow.left {  
  
    left: 26px;  
  
    transform: rotate(-6deg);  
  
}
```

```
.male-brow.right {  
  
    right: 26px;  
  
    transform: rotate(6deg);  
  
}
```

```
.male-collar {  
  
    position: absolute;  
  
    top: 120px;  
  
    left: 50%;  
  
    width: 152px;  
  
    height: 42px;  
  
    transform: translateX(-50%);  
  
    background: #eef7ff;
```



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```
border-radius: 0 0 26px 26px;

box-shadow: inset 0 4px 0 rgba(17, 82, 136,
0.08);

z-index: 1;

}
```

```
.male-collar::before,
.male-collar::after {

content: "";

position: absolute;

width: 52px;

height: 28px;

background: #eef7ff;

top: 0;

border-radius: 0 0 24px 24px;

border: 1px solid rgba(17, 82, 136, 0.12);

}
```

```
.male-collar::before {

left: -8px;
```

```
transform: rotate(12deg);  
  
clip-path: polygon(0 0, 100% 0, 100% 100%, 0  
70%);  
  
}
```

```
.male-collar::after {  
  
    right: -8px;  
  
    transform: rotate(-12deg);  
  
    clip-path: polygon(0 0, 100% 0, 100% 70%, 0  
100%);  
  
}
```

```
.male-shirt {  
  
    position: absolute;  
  
    bottom: 0;  
  
    left: 50%;  
  
    width: 170px;  
  
    height: 110px;  
  
    transform: translateX(-50%);  
  
    background: linear-gradient(180deg,  
#2e6c9a, #1f4f78);
```



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```
border-radius: 24px 24px 18px 18px;  
  
box-shadow: 0 16px 30px rgba(17, 82, 136,  
0.18);  
  
overflow: hidden;  
  
}
```

```
.male-shirt::before {  
  
  content: "";  
  
  position: absolute;  
  
  top: 14px;  
  
  left: 50%;  
  
  width: 86px;  
  
  height: 34px;  
  
  transform: translateX(-50%);  
  
  background: #eef7ff;  
  
  border-radius: 18px 18px 12px 12px;  
  
  z-index: 2;  
  
}
```

```
.male-shirt::after {
```

```
content: "";  
  
position: absolute;  
  
top: 42px;  
  
left: 50%;  
  
width: 20px;  
  
height: 44px;  
  
transform: translateX(-50%);  
  
background: #1f4f78;  
  
border-radius: 12px 12px 0 0;  
  
z-index: 2;  
  
}
```

```
.male-tie {  
  
    position: absolute;  
  
    top: 58px;  
  
    left: 50%;  
  
    width: 24px;  
  
    height: 38px;  
  
    transform: translateX(-50%);
```



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background: #e05b3a;

clip-path: polygon(50% 0%, 100% 40%, 60%
100%, 40% 100%, 0% 40%);

z-index: 3;

}

.profile-details h2 {

margin: 0;

font-size: 1.75rem;

color: var(--blue);

}

.profile-role {

margin: 10px 0 0;

color: #3a506b;

font-size: 1rem;

font-weight: 700;

letter-spacing: 0.02em;

}

```
.profile-details p {  
  
    margin: 8px 0 0;  
  
    color: var(--muted);  
  
    line-height: 1.75;  
  
}
```

```
.button-back {  
  
    display: inline-flex;  
  
    align-items: center;  
  
    justify-content: center;  
  
    padding: 14px 24px;  
  
    border-radius: 999px;  
  
    background: var(--blue);  
  
    color: #fff;  
  
    text-decoration: none;  
  
    font-weight: 700;  
  
    transition: transform 0.2s ease, box-shadow  
0.2s ease;  
  
}
```



```
.button-back:hover {  
  
    transform: translateY(-2px);  
  
    box-shadow: 0 18px 36px rgba(17, 82, 136,  
0.18);  
  
}  
  
.toast {  
  
    position: fixed;  
  
    right: 24px;  
  
    bottom: 24px;  
  
    max-width: 320px;  
  
    padding: 14px 18px;  
  
    border-radius: 14px;  
  
    background: rgba(15, 23, 42, 0.94);  
  
    color: #fff;  
  
    font-size: 0.95rem;  
  
    box-shadow: 0 18px 40px rgba(0, 0, 0, 0.24);  
  
    opacity: 0;  
  
    transform: translateY(12px);  
  
    pointer-events: none;
```


transition: opacity 0.25s ease, transform
0.25s ease;

z-index: 9999;

}

.toast.show {

opacity: 1;

transform: translateY(0);

}

@media (max-width: 720px) {

header {

align-items: flex-start;

}

.details th {

width: auto;

}

}

</style>

</head>

<body class="student-page">

<div class="page-wrapper">

<header>

<div class="student-heading">

<div>

<h1>Student Information</h1>

<p>Quick profile details for the student enrolled in the Computer Science and Engineering department.</p>

</div>

</div>

← Back to Department

</header>

<section class="profile-card">

<div class="male-avatar">

```
<div class="male-head">  
  <span class="male-brow left"></span>  
  <span class="male-brow right"></span>  
  <span class="male-eye left"></span>  
  <span class="male-eye right"></span>  
  <span class="male-nose"></span>  
  <span class="male-mouth"></span>  
</div>
```

```
<div class="male-collar"></div>  
<div class="male-shirt"></div>  
<div class="male-tie"></div>  
</div>
```

```
<div class="profile-details">  
  <h2>Ayush Katekhaye</h2>  
  <p class="profile-role">Full Stack Java  
Developer</p>  
  
  <p>5th Semester | Computer Science and  
Engineering student at Symbiosis Institute of  
Technology, Nagpur.</p>  
</div>
```

```
</section>
```



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```
<section class="card">
```

```
<table class="details">
```

```
<tr>
```

```
<th>Name</th>
```

```
<td>Ayush Katekhaye</td>
```

```
</tr>
```

```
<tr>
```

```
<th>PRN</th>
```

```
<td>24070521161</td>
```

```
</tr>
```

```
<tr>
```

```
<th>Department</th>
```

```
<td>Computer Science and  
Engineering</td>
```

```
</tr>
```

```
<tr>
```

```
<th>Semester</th>
```

```
<td>5th Semester</td>
```

```
</tr>
```

```
<tr>

    <th>College</th>

    <td>Symbiosis Institute of Technology,
Nagpur</td>

</tr>

</table>

</section>

</div>

<div id="toast" class="toast"></div>

<script src="script.js"></script>

</body>

</html>

Script.js

window.addEventListener("DOMContentLoaded", function () {

    const body = document.body;

    const toast =
document.getElementById("toast");

    let toastTimer = null;

    if (body.classList.contains("department-page"))
```

```
&& !sessionStorage.getItem("welcomeShown"))  
{  
    alert("Welcome to SIT NAGPUR");  
    sessionStorage.setItem("welcomeShown",  
"1");  
}
```

```
function showToast(message) {  
    if (!toast) return;  
    toast.textContent = message;  
    toast.classList.add("show");  
    clearTimeout(toastTimer);  
    toastTimer = setTimeout(() =>  
toast.classList.remove("show"), 1800);  
}
```

```
document.addEventListener("click", function  
(event) {  
    const button =  
event.target.closest(".clickable");  
    if (!button) return;
```

```
const message = button.dataset.toast;

if (!message) return;

event.preventDefault();

showToast(message);

const href = button.getAttribute("href");
if (href) {
    if (href.startsWith("#")) {
        window.location.hash = href;
    } else {
        setTimeout(() => {
            window.location.href = href;
        }, 250);
    }
}

});

});
```



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// Function to open the Student Information
page

```
function openStudentPage() {  
    window.location.href = "student.html";  
}
```

// Function to return to the Department page

```
function goBack() {  
    window.location.href = "department.html";  
}
```


SIT Nagpur
Logo

Symbiosis Institute of Technology

Computer Science and Engineering Department

Home

Highlights

Student Info

Build the future with AI, Cloud, Data Science and Cyber Security.

The Department of Computer Science and Engineering at SIT Nagpur empowers students with hands-on learning, modern labs and industry-aligned projects for careers in Programming, Artificial Intelligence, Data Science, Cyber Security, Cloud Computing and Software Engineering.

View Student Information

Explore Highlights



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SIT Nagpur
Logo

Symbiosis Institute of Technology

Computer Science and Engineering Department

Home

Highlights

Student Info

Practical Learning

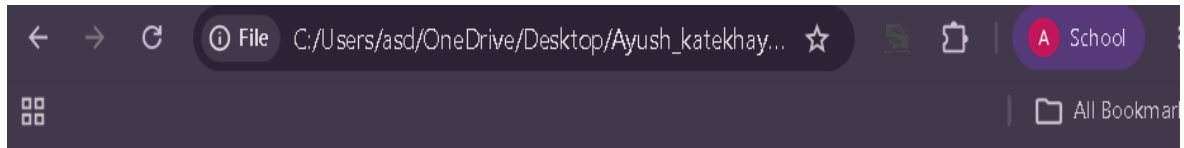
Project-based labs and campus events give students real experience solving industry problems.

Modern Technology

Students work with AI, cloud platforms, cybersecurity tooling and full-stack software systems.

Student Success

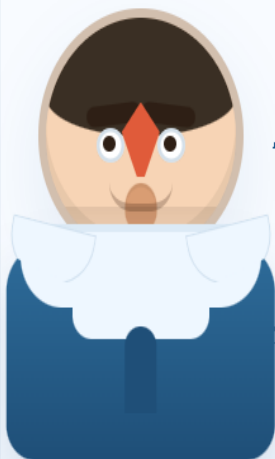
Strong placement support, mentoring and a culture of innovation help learners achieve their goals.



Student Information

Quick profile details for the student enrolled in the Computer Science and Engineering department.

[← Back to Department](#)



Ayush Katekhaye

Full Stack Java Developer

5th Semester | Computer Science and Engineering student at
Symbiosis Institute of Technology, Nagpur.

Name Ayush Katekhaye

PRN 24070521161

Department Computer Science and Engineering



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127.0.0.1:5500/student.html



School



All Bookmarks



Ayush Katekhaye

Full Stack Java Developer

5th Semester | Computer Science and Engineering student at
Symbiosis Institute of Technology, Nagpur.

Name

Ayush Katekhaye

PRN

24070521161

Department

Computer Science and Engineering

Semester

5th Semester

College

Symbiosis Institute of Technology, Nagpur

Practical 2

Aim: Use var, let, const, and demonstrate basic features like template literals and destructuring; build a billing calculator with user input.

Software/ Tools Required: vs code , web browser

Experiment program code:

Index.html

```
<!DOCTYPE html>

<html>

<head>

  <title>Krushi Kendra Receipt</title>

<style>

  body{

    font-family: Arial, sans-serif;

    background:#f4f4f4;

  }

  .receipt{

    width:450px;

    margin:30px auto;

    background:white;

    border:2px solid black;

    padding:20px;
```



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```
    box-shadow:0 0 10px gray;
}
```

```
h2,h3{
    text-align:center;
    margin:5px;
}
```

```
hr{
    border:1px dashed black;
}
```

```
table{
    width:100%;
    border-collapse:collapse;
    margin-top:10px;
}
```

```
table,th,td{
    border:1px solid black;
}
```

```
th,td{
```



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padding:8px;

text-align:center;

}

.total{

text-align:right;

margin-top:15px;

font-size:18px;

font-weight:bold;

}

.footer{

text-align:center;

margin-top:20px;

}

</style>

</head>

<body>

<script>

const GST = 0.18;

```
var customerName = prompt("Enter Customer Name:");

var productName = prompt("Enter Product Name:");

var quantity = parseInt(prompt("Enter Quantity:"));

var price = parseFloat(prompt("Enter Price per Item:"));


let total = quantity * price;

let gst = total * GST;

let finalCost = total + gst;


document.write(`

<div class="receipt">

<h2> 🌿 jai laxmi KRUSHI KENDRA 🌿 </h2>

<h3>Chikhali, Adyal</h3>

<p style="text-align:center;">Mob : 9876543210</p>

<hr>

<p><b>Customer :</b> ${customerName}</p>

<p><b>Date :</b> ${new Date().toLocaleDateString()}</p>
```


<table>

<tr>

<th>Product</th>

<th>Qty</th>

<th>Rate</th>

<th>Total</th>

</tr>

<tr>

<td>\${productName}</td>

<td>\${quantity}</td>

<td>₹\${price.toFixed(2)}</td>

<td>₹\${total.toFixed(2)}</td>

</tr>

</table>

<p class="total">GST (18%) : ₹\${gst.toFixed(2)}</p>

<p class="total">Final Amount : ₹\${finalCost.toFixed(2)}</p>

<hr>

```
<div class="footer">
```

```
<h3>Thank You... Visit Again!</h3>
```

```
<p>Ayush Katekhaye</p>
```

```
<p>PRN No : 24070521161</p>
```

```
</div>
```

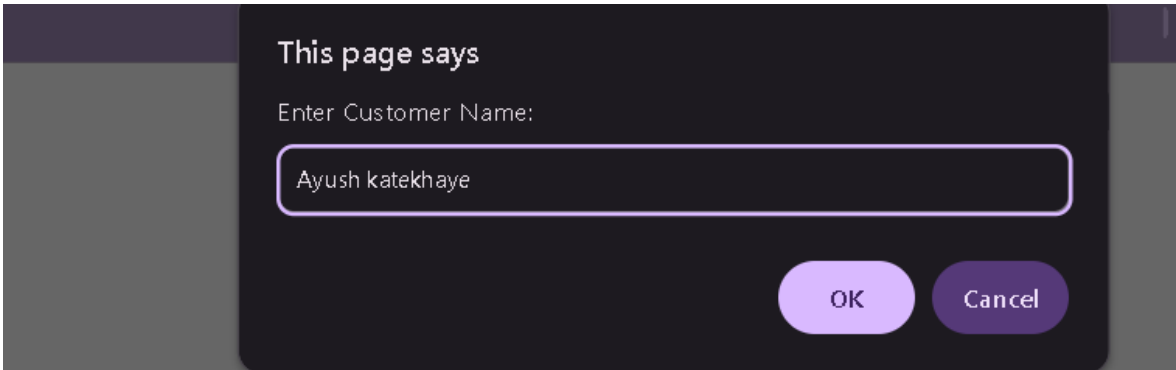
```
</div>
```

```
`);
```

```
</script>
```

```
</body>
```

Output:



The screenshot shows a dark-themed web interface. At the top, a dark purple header bar contains the text "This page says" in white. Below this, the text "Enter Customer Name:" is displayed in a light gray font. A white text input field with a light gray border contains the text "Ayush katekhaye". At the bottom right of the form, there are two rounded rectangular buttons: a light purple "OK" button and a dark purple "Cancel" button.



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This page says

Enter Product Name:

Urea

OK

Cancel

This page says

Enter Quantity:

1

OK

Cancel

This page says

Enter Price per Item:

1400

OK

Cancel



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File C:/Users/asd/OneDrive/Desktop/Ayush_kat...



A School



All Bookmarks

 **jai laxmi KRUSHI KENDRA** 

Chikhali, Adyal

Mob : 9876543210

Customer : ayush katekhaye

Date : 8/7/2026

Product	Qty	Rate	Total
Urea	1	₹1400.00	₹1400.00

GST (18%) : ₹252.00

Final Amount : ₹1652.00

Thank You... Visit Again!

Ayush Katekhaye

PRN No : 24070521161

Case study 2

Case Study Title:- Develop the webpage for an online shopping website with using HTML,CSS and HTML5 tags

Software/Tools Used:- VScode , Web Browser

Code:

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width,
initial-scale=1.0">

    <title>Green Basket - Online Vegetable Shopping</title>

    <link rel="stylesheet" href="style.css">

</head>

<body>

<header>
```

<h1> 🌿 Green Basket</h1>

<h2>Ayush Katehay</h2>

<h2>24070521161</h2>

<p>Fresh Vegetables Delivered to Your Doorstep</p>

</header>

<nav>

Home

Vegetables

Offers

Contact

</nav>

<div class="container">

<h2>Fresh Vegetables</h2>

<div class="products">

<div class="product">

<h3>Tomato</h3>

<p>₹40 / Kg</p>

</div>

<div class="product">

<h3>Potato</h3>

<p>₹30 / Kg</p>

</div>

<div class="product">

<h3>Onion</h3>

<p>₹35 / Kg</p>

</div>

</div>

<hr>

<h2>Customer Billing</h2>

<form>

<label>Customer Name</label>

<input type="text" id="name">

<label>Mobile Number</label>

<input type="tel" id="mobile">

<label>Invoice Number</label>

<input type="text" id="invoice">

<label>Select Product</label>

<select id="product">

<option value="Tomato">Tomato</option>

<option value="Potato">Potato</option>

<option value="Onion">Onion</option>

</select>

<label>Quantity (Kg)</label>

<input type="number" id="qty">

<label>Rate per Kg (₹)</label>

<input type="number" id="rate" readonly>

<label>Discount (%)</label>

<input type="number" id="discount">

<label>GST (%)</label>

<input type="number" id="gst">

<label>Packing Charges (₹)</label>

<input type="number" id="packing">

<label>Payment Mode</label>

<select id="payment">

<option>Cash</option>

<option>Card</option>

<option>UPI</option>

</select>

<label>Membership</label>

<select id="member">

<option>Yes</option>

<option>No</option>

</select>

<button type="button" onclick="calculateBill()">

Calculate Bill

</button>

</form>

<div id="bill"></div>



</div>

<footer>

<p>© 2026 Green Basket | Fresh Vegetable Shopping</p>

</footer>

<script src="script.js"></script>

</body>

</html>

style.css

```
*{  
  
    margin:0;  
  
    padding:0;  
  
    box-sizing:border-box;  
  
    font-family:Arial, Helvetica, sans-serif;  
  
}
```

```
body{  
  
    background:#f2f8f2;
```



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```
}
```

```
/* Header */
```

```
header{  
  
    background:#2e7d32;  
  
    color:white;  
  
    text-align:center;  
  
    padding:20px;  
  
}
```

```
header h1{  
  
    font-size:40px;  
  
}
```

```
header p{  
  
    margin-top:8px;  
  
}
```

```
/* Navigation */
```

```
nav{  
  
    background:#388e3c;
```



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```
text-align:center;

padding:12px;

}
```

```
nav a{

    color:white;

    text-decoration:none;

    margin:20px;

    font-size:18px;

    font-weight:bold;

}
```

```
nav a:hover{

    color:yellow;

}
```

```
/* Container */
```

```
.container{

    width:80%;

    margin:30px auto;

    background:white;
```

```
padding:20px;  
  
border-radius:10px;  
  
box-shadow:0px 0px 10px gray;  
  
}
```

```
/* Product Cards */
```

```
.products{  
  
    display:flex;  
  
    justify-content:space-around;  
  
    flex-wrap:wrap;  
  
    gap:20px;  
  
    margin:20px 0;  
  
}
```

```
.product{  
  
    width:200px;  
  
    border:1px solid #ddd;  
  
    border-radius:10px;
```



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text-align:center;

padding:15px;

box-shadow:0px 0px 8px lightgray;

}

.product img{

width:160px;

height:160px;

object-fit:cover;

border-radius:10px;

}

.product h3{

color:green;

margin-top:10px;

}

.product p{

font-weight:bold;

}

/* Form */

label{

display:block;

margin-top:15px;

font-weight:bold;

}

input,select{

width:100%;

padding:10px;

margin-top:5px;

border-radius:5px;

border:1px solid gray;



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```
}
```

```
button{
```

```
    margin-top:20px;
```

```
    width:100%;
```

```
    padding:12px;
```

```
    background:green;
```

```
    color:white;
```

```
    border:none;
```

```
    font-size:18px;
```

```
    cursor:pointer;
```

```
    border-radius:5px;
```

```
}
```

```
button:hover{
```

```
    background:#1b5e20;
```

```
}
```

```
/* Invoice */
```

```
#bill{  
  
    margin-top:25px;  
  
    padding:20px;  
  
    background:#ecf0f1;  
  
    border-radius:10px;  
  
}
```

```
/* Footer */
```

```
footer{  
  
    margin-top:30px;  
  
    background:#2e7d32;  
  
    color:white;  
  
    text-align:center;  
  
    padding:15px;  
  
}
```

script.js

```
// Vegetable Prices
```

```
const rates = {
```

```
    Tomato:40,
```

```
    Potato:30,
```

```
    Onion:35
```

```
};
```

```
// Automatic Price
```

```
document.getElementById("product").addEventListener("change",function(){
```

```
document.getElementById("rate").value=rates[this.value];
```

```
});
```

```
// Default Rate
```

```
window.onload=function(){
```

```
document.getElementById("rate").value=40;
```

```
};
```

```
// Calculate Bill
```

```
function calculateBill(){
```

```
let  
name=document.getElementById("name").value  
;
```

```
let  
mobile=document.getElementById("mobile").val
```



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ue;

let

invoice=document.getElementById("invoice").value;

let

product=document.getElementById("product").value;

let

qty=parseFloat(document.getElementById("qty").value);

let

rate=parseFloat(document.getElementById("rate").value);

let

discount=parseFloat(document.getElementById("discount").value)||0;

let

gst=parseFloat(document.getElementById("gst"))

```
.value)||0;
```

```
let  
packing=parseFloat(document.getElementById("  
packing").value)||0;
```

```
let  
payment=document.getElementById("payment"  
).value;
```

```
let  
member=document.getElementById("member")  
.value;
```

```
// Amount
```

```
let amount=qty*rate;
```

```
// Discount
```

```
let discountAmount=(amount*discount)/100;
```

```
let subtotal=amount-discountAmount;
```

```
// Membership Discount
```

```
if(member=="Yes"){
```

```
    subtotal=subtotal-(subtotal*5/100);
```

```
}
```

```
// GST
```

```
let gstAmount=(subtotal*gst)/100;
```

```
// Final Bill
```

```
let finalBill=subtotal+gstAmount+packing;
```

```
// Display Invoice
```


document.getElementById("bill").innerHTML=`

<h2 align="center">Invoice</h2>

<hr>

<p>Customer : \${name}</p>

<p>Mobile : \${mobile}</p>

<p>Invoice No : \${invoice}</p>

<p>Product : \${product}</p>

<p>Quantity : \${qty} Kg</p>

<p>Rate : ₹\${rate}</p>

<p>Total : ₹\${amount.toFixed(2)}</p>

<p>Discount :

₹\${discountAmount.toFixed(2)}</p>

<p>GST : ₹\${gstAmount.toFixed(2)}</p>

<p>Packing : ₹\${packing.toFixed(2)}</p>

<p>Payment : \${payment}</p>

<p>Membership : \${member}</p>

<hr>

<h2 style="color:green">

Final Bill : ₹\${finalBill.toFixed(2)}

</h2>



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;

}

File

C:/Users/asd/OneDrive/Desktop/Ayush_katekhay...

☆

School

All Bookmarks



Green Basket

Ayush Katehaye
24070521161

Fresh Vegetables Delivered to Your Doorstep

Home

Vegetables

Offers

Contact

Fresh Vegetables



Tomato
₹40 / Kg



Potato
₹30 / Kg





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Onion

₹35 / Kg

Customer Billing

Customer Name

Mobile Number

Invoice Number

Select Product

Quantity (Kg)



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←

→

↺

File

C:/Users/asd/OneDrive/Desktop/Ayush_katekhay...

☆

A School

⋮

☐☐☐☐

All Bookmarks

Rate per Kg (₹)

Discount (%)

GST (%)

Packing Charges (₹)

Payment Mode

Cash

▼

Membership

Yes

▼

Calculate Bill

© 2026 Green Basket | Fresh Vegetable Shopping

Practical No 3

Aim: Implement control structures and form validation; create a grading system based on user-entered marks.

Software/Tools :Vs code , chrome browser

Experiment program code

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport"
content="width=device-width, initial-
scale=1.0">

<title>Student Grading System</title>


<link rel="stylesheet" href="style.css">


</head>

<body>

<div class="container">
```

<h1> 🎓 Student Grading System</h1>

<label>Student Name</label>

<input type="text" id="name"
placeholder="Enter Student Name">

<label>Student PRN</label>

<input type="text" id="prn" placeholder="Enter
Student PRN">

<label>English</label>

<input type="number" id="english"
placeholder="Enter English Marks">

<label>Mathematics</label>

<input type="number" id="maths"
placeholder="Enter Mathematics Marks">

<label>Science</label>

<input type="number" id="science"
placeholder="Enter Science Marks">

<label>Computer</label>

<input type="number" id="computer"
placeholder="Enter Computer Marks">

<label>History</label>

<input type="number" id="history"
placeholder="Enter History Marks">

<button onclick="calculateGrade()">

Calculate Result

</button>

<div id="result"></div>

</div>

<script src="script.js"></script>

</body>

</html>

Style.css

{

margin:0;

padding:0;

box-sizing:border-box;

font-family:Arial,Helvetica,sans-serif;

}

body{

background:linear-gradient(135deg,#4facfe,#00f2fe);

display:flex;

justify-content:center;

align-items:center;



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min-height:100vh;

padding:20px;

}

.container{

width:450px;

background:white;

padding:30px;

border-radius:20px;

box-shadow:0 10px 30px rgba(0,0,0,.3);

}

h1{

text-align:center;

margin-bottom:25px;

color:#1565c0;

}

label{

display:block;

margin-top:15px;

font-weight:bold;



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color:#333;

}

input{

width:100%;

padding:12px;

margin-top:6px;

border:2px solid #ddd;

border-radius:10px;

font-size:16px;

transition:.3s;

```
}
```

```
input:focus{
```

```
border-color:#1565c0;
```

```
outline:none;
```

```
}
```

```
button{
```

```
width:100%;
```

```
margin-top:25px;
```

```
padding:15px;
```



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border:none;

border-radius:10px;

background:#1565c0;

color:white;

font-size:18px;

cursor:pointer;

transition:.3s;

}

button:hover{

background:#0d47a1;

transform:scale(1.02);

}

#result{

display:none;

margin-top:25px;

background:#f8f9fa;

padding:20px;

border-radius:10px;

border-left:6px solid #1565c0;



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line-height:1.8;

font-size:17px;

}

.success{

color:green;

font-weight:bold;

}

.fail{

color:red;

font-weight:bold;


```
}
```

Script.js

```
function calculateGrade() {  
  
    let  
    name=document.getElementById("name").value  
    .trim();  
  
    let prn =  
    document.getElementById("prn").value.trim();  
  
    let  
    english=Number(document.getElementById("en  
    glish").value);  
  
    let  
    maths=Number(document.getElementById("mat  
    hs").value);  
  
    let  
    science=Number(document.getElementById("sci  
    ence").value);  
  
    let
```

```
computer=Number(document.getElementById("computer").value);
```

```
let  
history=Number(document.getElementById("history").value);
```

```
// Name Validation
```

```
if(name=="")  
{  
    alert("Please enter student name.");  
    return;  
}
```

```
// PRN Validation
```

```
if(prn=="")  
{  
    alert("Please enter student PRN.");  
    return;
```

```
}
```

```
if(!/^\\d{11}$/.test(prn))
```

```
{
```

```
    alert("PRN must contain exactly 11 digits.");
```

```
    return;
```

```
}
```

```
// Marks Validation
```

```
let
```

```
subjects=[english,maths,science,computer,histo  
ry];
```

```
for(let i=0;i<subjects.length;i++)
```

```
{
```

```
    if(isNaN(subjects[i]) || subjects[i]<0 ||  
    subjects[i]>100)
```

```
{
```

```
    alert("Enter valid marks (0-100) for all  
    subjects.");
```



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```
return;
```

```
}
```

```
}
```

```
// Total
```

```
let
```

```
total=english+maths+science+computer+history;
```

```
// Percentage
```

```
let percentage=total/5;
```

```
// Grade
```

```
let grade="";
```

```
let remark="";
```

```
if(percentage>=90)
```

```
{  
  
grade="A+";  
remark="Outstanding";  
  
}  
  
else if(percentage>=80)  
  
{  
  
grade="A";  
remark="Excellent";  
  
}  
  
else if(percentage>=70)  
  
{  
  
grade="B";  
remark="Very Good";  
  
}  
  
else if(percentage>=60)  
  
{  
  
grade="C";  
remark="Good";  
  
}
```



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```
else if(percentage>=50)
```

```
{
```

```
grade="D";
```

```
remark="Average";
```

```
}
```

```
else
```

```
{
```

```
grade="F";
```

```
remark="Fail";
```

```
}
```

```
// Pass or Fail
```

```
let status;
```

```
if(
```

```
english>=35 &&
```

```
maths>=35 &&
```

```
science>=35 &&
```

```
computer>=35 &&

history>=35

)

{

status="<span class='success'>PASS
✓</span>";

}

else

{

status="<span class='fail'>FAIL ✗</span>";

}


// Display Result

let result=document.getElementById("result");

result.style.display="block";

result.innerHTML=`
```

Student Report Card

 Student Name : \${name}

 English : \${english}

 Mathematics : \${maths}

 Science : \${science}

 Computer : \${computer}

 History : \${history}

  Total Marks : $\frac{\text{\$total}}{500}$

  Percentage :
 $\text{\$percentage.toFixed(2)} \%$

  Grade : $\text{\$grade}$

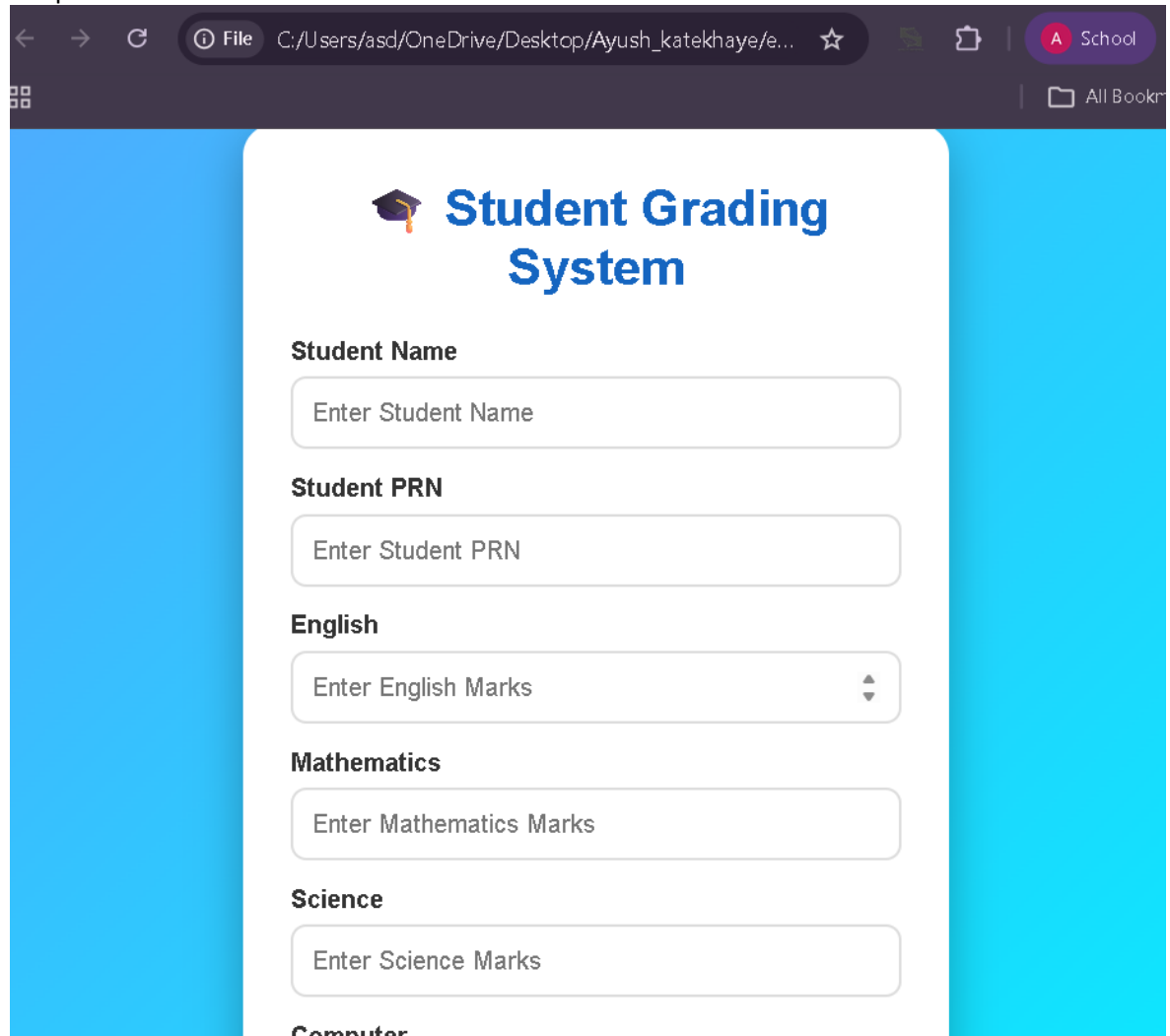
  Remark : $\text{\$remark}$

  Result : $\text{\$status}$

`
;

}

Output:



The screenshot shows a web browser window with a dark theme. The address bar displays the file path: C:/Users/asd/OneDrive/Desktop/Ayush_katekhaye/e... The browser has several tabs, with the active one labeled 'School'. The main content area features a 'Student Grading System' form with a blue header and a white body. The form includes input fields for Student Name, Student PRN, English Marks, Mathematics Marks, Science Marks, and Computer Marks. The 'English' input field is a spinner control.

 **Student Grading System**

Student Name

Student PRN

English

Mathematics

Science

Computer



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File

C:/Users/asd/OneDrive/Desktop/Ayush_katekhaye/e...



A

School



All Bookmarks

English

Mathematics

Science

Computer

History

Calculate Result



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Calculate Result

Student Report Card

 **Student Name :** AYUSH KATEKHAYE

 **English :** 100

 **Mathematics :** 96

 **Science :** 79

 **Computer :** 90

 **History :** 78

 **Total Marks :** 443 / 500

 **Percentage :** 88.60 %

 **Grade :** A

 **Remark :** Excellent

 **Result :** **PASS** 



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Calculate Result

Student Report Card

 **Student Name :** AYUSH KATEKHAYE

 **English :** 100

 **Mathematics :** 96

 **Science :** 79

 **Computer :** 90

 **History :** 78

 **Total Marks :** 443 / 500

 **Percentage :** 88.60 %

 **Grade :** A

 **Remark :** Excellent

 **Result :** **PASS** 

Case study 3

Aim: Scholarship test

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport"
content="width=device-width, initial-
scale=1.0">

<title>Login & Student Grade System</title>

<link rel="stylesheet" href="style.css">

</head>

<body>

<div class="container">

<!-- Login Section -->

<div id="loginSection">
```

<h1>🔒 Login Page</h1>

<label>Username</label>

<input type="text" id="username"
placeholder="Enter Username">

<label>Password</label>

<input type="password" id="password"
placeholder="Enter Password">

<button
onclick="validateLogin()">Login</button>

</div>

<!-- Grade Section -->

<div id="gradeSection" style="display:none;">

<h1> 🎓 Student Grade Sheet</h1>

<label>Student Name</label>

<input type="text" id="studentName"
placeholder="Enter Student Name">

<label>English</label>

<input type="number" id="english">

<label>Mathematics</label>

<input type="number" id="maths">

<label>Science</label>

<input type="number" id="science">

<label>Computer</label>

<input type="number" id="computer">

<label>History</label>

<input type="number" id="history">

```
<button onclick="calculateGrade()">Calculate  
Grade</button>
```

```
<div id="result"></div>
```

```
</div>
```

```
</div>
```

```
<script src="script.js"></script>
```

```
</body>
```

```
</html>
```

Style.css

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<meta name="viewport"
```

```
content="width=device-width, initial-  
scale=1.0">
```

```
<title>Login & Student Grade System</title>
```

```
<link rel="stylesheet" href="style.css">
```

```
</head>
```

```
<body>
```

```
<div class="container">
```

```
<!-- Login Section -->
```

```
<div id="loginSection">
```

```
<h1>🔒 Login Page</h1>
```

```
<label>Username</label>
```

```
<input type="text" id="username"  
placeholder="Enter Username">
```

<label>Password</label>

<input type="password" id="password"
placeholder="Enter Password">

<button
onclick="validateLogin()">Login</button>

</div>

<!-- Grade Section -->

<div id="gradeSection" style="display:none;">

<h1>🎓 Student Grade Sheet</h1>

<label>Student Name</label>

<input type="text" id="studentName"
placeholder="Enter Student Name">

<label>English</label>

<input type="number" id="english">

<label>Mathematics</label>

<input type="number" id="maths">

<label>Science</label>

<input type="number" id="science">

<label>Computer</label>

<input type="number" id="computer">

<label>History</label>

<input type="number" id="history">

<button onclick="calculateGrade()">Calculate
Grade</button>

<div id="result"></div>

</div>



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```
</div>
```

```
<script src="script.js"></script>
```

```
</body>
```

```
</html>
```

Script.js

```
function validateLogin()
```

```
{
```

```
let
```

```
username=document.getElementById("username").value.trim();
```

```
let
```

```
password=document.getElementById("password").value;
```

```
// Username Validation
```

```
if(username=="")
```

```
{  
  
alert("Please Enter Username");  
  
return;  
  
}  
  
// Password Validation  
  
if(password=="")  
{  
  
alert("Please Enter Password");  
  
return;  
  
}  
  
if(password.length<8)  
{  
  
alert("Password must contain at least 8  
characters.");  
  
return;  
  
}
```



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```
let upper=/[A-Z]/;
```

```
let lower=/[a-z]/;
```

```
if(!upper.test(password))
```

```
{
```

```
    alert("Password must contain at least one  
    Uppercase Letter.");
```

```
    return;
```

```
}
```

```
if(!lower.test(password))
```

```
{
```

```
    alert("Password must contain at least one  
    Lowercase Letter.");
```

```
    return;
```

```
}
```

```
alert("Login Successful");
```

```
document.getElementById("loginSection").style.  
display="none";
```



```
document.getElementById("gradeSection").style  
.display="block";
```

```
}
```

```
function calculateGrade()
```

```
{
```

```
let
```

```
student=document.getElementById("studentNa  
me").value.trim();
```

```
let
```

```
english=Number(document.getElementById("en  
glish").value);
```

```
let
```

```
maths=Number(document.getElementById("mat  
hs").value);
```

```
let
```

```
science=Number(document.getElementById("sci  
ence").value);
```

```
let
```

```
computer=Number(document.getElementById("c  
omputer").value);
```



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```
let  
history=Number(document.getElementById("his  
tory").value);
```

```
if(student=="")  
{  
alert("Please Enter Student Name");  
return;  
}
```

```
let  
subjects=[english,maths,science,computer,histo  
ry];
```

```
for(let i=0;i<subjects.length;i++)  
{  
if(isNaN(subjects[i]) || subjects[i]<0 ||  
subjects[i]>100)  
{  
alert("Enter valid marks (0-100) for all  
subjects.");  
return;
```

}

}

```
let  
total=english+maths+science+computer+history;  
  
let percentage=total/5;
```

```
let grade="";
```

```
let remark="";
```

```
if((percentage>=50 && percentage<=65)
```

```
{
```

```
    grade="C";
```

```
    remark="Good";
```

```
}
```

```
else if((percentage>=66 && percentage<=80)
```

```
{
```

```
    grade="B";
```

```
    remark="Very Good";
```

```
}
```



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```
else if(percentage>=81 && percentage<=90)
```

```
{
```

```
grade="A";
```

```
remark="Excellent";
```

```
}
```

```
else if(percentage>90)
```

```
{
```

```
grade="A+";
```

```
remark="Outstanding";
```

```
}
```

```
else
```

```
{
```

```
grade="Fail";
```

```
remark="Need Improvement";
```

```
}
```

```
let status;
```

```
if(
```

english>=35 &&

maths>=35 &&

science>=35 &&

computer>=35 &&

history>=35

)

{

status="PASS

✓";

}

else

{

status="FAIL ✗";

}

document.getElementById("result").style.display
="block";

document.getElementById("result").innerHTML
=

Student Report Card

 Student Name : \${student}

 English : \${english}

 Mathematics : \${maths}

 Science : \${science}

 Computer : \${computer}

 History : \${history}

 Total : \${total} / 500

 Percentage :

`${percentage.toFixed(2)} %<br>`

`<b> 🏆 Grade :</b> ${grade}<br>`

`<b> 💬 Remark :</b> ${remark}<br>`

`<b> 📌 Result :</b> ${status}`

``;`

`}`



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File

C:/Users/asd/OneDrive/Desktop/Ayush_katekhaye/C...

School

All Bookm

This page says
Login Successful
OK

 Login Page

Username

24070521161

Password

.....

Login

Experiment no 4

Aim: Use function types, scope, and closures; apply try-catch; build a palindrome checker.

Software/Tools Required: Vs code , Chrome browser

Palindrome check

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport"
content="width=device-width, initial-
scale=1.0">

    <title>Palindrome Checker</title>

    <link rel="stylesheet" href="style.css">

</head>

<body>

    <div class="container">

        <h1>Ayush Katekhaye</h1>
```

<h1>24070521161</h1>

<h1>Palindrome Checker</h1>

<input

type="text"

id="textInput"

placeholder="Enter a word or sentence">

<button onclick="checkPalindrome()">

Check

</button>

<div id="result"></div>

<div class="concepts">

<h3>JavaScript Concepts Used</h3>

✓ Function Declaration

✓ Function Expression

✓ Arrow Function

✓ Global Scope

✓ Local Scope

✓ Closures

✓ Try-Catch

</div>

</div>

<script src="script.js"></script>

</body>

</html>



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Style.css

```
*{  
  
    margin:0;  
  
    padding:0;  
  
    box-sizing:border-box;  
  
    font-family:Arial, Helvetica, sans-serif;  
  
}
```

```
body{  
  
  
  
  
    height:100vh;  
  
  
  
    display:flex;  
  
    justify-content:center;  
  
    align-items:center;  
  
  
    background:linear-  
gradient(135deg,#4facfe,#00f2fe);
```

}

.container{

width:420px;

background:white;

padding:30px;

border-radius:15px;

box-shadow:0 10px 25px rgba(0,0,0,.25);

}

h1{

text-align:center;



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margin-bottom:20px;

color:#333;

}

input{

width:100%;

padding:12px;

border-radius:8px;

border:1px solid #ccc;

font-size:16px;

margin-bottom:15px;

}

button{

width:100%;

padding:12px;

border:none;

background:#007BFF;

color:white;

font-size:16px;

border-radius:8px;



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cursor:pointer;

}

button:hover{

background:#0056b3;

}

#result{

margin-top:20px;

text-align:center;

font-size:18px;

font-weight:bold;

}

.concepts{

margin-top:25px;

}

.concepts h3{

color:#444;

margin-bottom:10px;

}



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```
.concepts ul{
```

```
padding-left:20px;
```

```
}
```

```
.concepts li{
```

```
margin:6px 0;
```

```
}
```

Script.js

```
// =====
```

```
// Global Scope
```

```
// =====
```

```
let appName = "Palindrome Checker";
```

```
// =====
```

```
// Closure Example
```

```
// =====
```

```
function createCounter() {
```

```
    let count = 0;
```

```
    return function () {
```

```
        count++;
```

```
        return count;
```

```
    };
```

```
}
```

```
const counter = createCounter();
```

```
// =====
```

```
// Function Declaration
```

```
// =====
```

```
function checkPalindrome() {
```

```
    try {
```

```
        let input =  
        document.getElementById("textInput").value;
```

```
        if (input.trim() === "") {
```

```
            throw "Please enter some text.";
```

```
        }
```

```
// =====
```

```
// Function Expression
```

```
// =====
```

```
const cleanText = function (text) {
```

```
    return text
```

```
        .toLowerCase()
```

```
        .replace(/[^a-z0-9]/g, "");
```

```
};
```

```
// =====
```

```
// Arrow Function
```

```
// =====
```

```
const reverseText = text =>
```

```
    text.split("").reverse().join("");
```

```
let cleaned = cleanText(input);
```

```
let result =
```

```
    cleaned === reverseText(cleaned);
```

```
let visits = counter();
```

```
document.getElementById("result").innerHTML +=  
TML =
```

```
    result
```

```
    ? `  "${input}" is a  
    Palindrome.<br><br>Checked ${visits} time(s).`
```

```
    : `  "${input}" is NOT a  
    Palindrome.<br><br>Checked ${visits} time(s).`;
```

```
    }
```

```
catch (error) {
```

document.getElementById("result").innerHTML
TML =

```
`<span  
style="color:red;">${error}</span>`;
```

```
}
```

```
}
```

Output:



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Ayush Katekhaye

24070521161

Palindrome Checker

wow

Check

✔ "wow" is a Palindrome.

Checked 1 time(s).

JavaScript Concepts Used

- ✔ Function Declaration
- ✔ Function Expression
- ✔ Arrow Function
- ✔ Global Scope
- ✔ Local Scope
- ✔ Closures



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Case sst

Case study 4

Aim: Real-Time Case Study: ATM Card PIN Verification Using JavaScript Functions

Case Study An ATM system verifies whether a customer's PIN is a palindrome. If the PIN reads the same forward and backward, the bank displays a special security message. To implement this, create a function to reverse the entered PIN number and another function to check whether it is a palindrome. Use function declaration, function expression, arrow function, and demonstrate scope concepts

Software/Tools Used:- VScode , Web Browser

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<meta name="viewport"  
content="width=device-width, initial-  
scale=1.0">
```

```
<title>ATM PIN Verification</title>
```

```
<link rel="stylesheet" href="style.css">
```

</head>

<body>

<div class="container">

<h2>Ayush Katekhaye</h2>

<h3>24070521161</h3>

<h1>ATM PIN Verification</h1>

<input type="password"

id="pin"

maxlength="6"

placeholder="Enter ATM PIN">

<button onclick="verifyPIN()">

Verify PIN

</button>

<div id="result"></div>

</div>

<script src="script.js"></script>

</body>

</html>

Style.css

```
*{  
  
    margin:0;  
  
    padding:0;  
  
    box-sizing:border-box;  
  
    font-family:Arial, Helvetica, sans-serif;  
  
}
```



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```
body{

    height:100vh;

    display:flex;

    justify-content:center;

    align-items:center;

    background:linear-
    gradient(135deg,#1e90ff,#00c6ff);

}

.container{

    width:430px;

    background:white;

    padding:35px;
```

border-radius:15px;

box-shadow:0 10px 25px rgba(0,0,0,.25);

text-align:center;

}

h1{

margin:20px 0;

color:#222;

}

h2,h3{



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color:#333;

margin-bottom:10px;

}

input{

width:100%;

padding:15px;

font-size:18px;

border-radius:8px;

border:1px solid gray;

margin-top:20px;


```
}
```

```
button{
```

```
width:100%;
```

```
margin-top:20px;
```

```
padding:15px;
```

```
font-size:18px;
```

```
border:none;
```

```
background:#1565c0;
```

```
color:white;
```



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border-radius:8px;

cursor:pointer;

}

button:hover{

background:#0d47a1;

}

#result{

margin-top:25px;

font-size:18px;

font-weight:bold;

}

Script.js

```
// =====
```

```
// Global Scope
```

```
// =====
```

```
let bankName = "ABC Bank";
```

```
// =====
```

```
// Function Expression
```

```
// Reverse PIN
```

```
// =====
```

```
const reversePIN = function(pin){
```

```
    return pin.split("").reverse().join("");
```



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```
};
```

```
// =====
```

```
// Arrow Function
```

```
// Check Palindrome
```

```
// =====
```

```
const isPalindrome = (pin)=>{
```

```
    return pin === reversePIN(pin);
```

```
};
```

```
// =====
```

```
// Function Declaration
```

```
// =====
```

```
function verifyPIN(){
```

```
// Local Scope
```

```
let pin =  
document.getElementById("pin").value;
```

```
let result =  
document.getElementById("result");
```

```
if(pin.trim()==="){
```

```
    result.innerHTML="✗ Please enter ATM  
PIN.";
```

```
    result.style.color="red";
```

```
    return;
```

```
}
```

```
if(!/^[0-9]+$/.test(pin)){
```

```
result.innerHTML=" ❌ PIN should contain  
only numbers.";
```

```
result.style.color="red";
```

```
return;
```

```
}
```

```
if(isPalindrome(pin)){
```

```
result.innerHTML=
```

```
"⚠️ Weak PIN!<br><br>" +
```

```
"Your PIN is a palindrome.<br>" +
```

```
"Example: 1221, 4554, 9889<br><br>" +
```

```
"Please change your PIN.";
```

```
result.style.color="red";
```

}

else{

result.innerHTML=

" Strong PIN!

" +

"Your PIN is NOT a palindrome.
" +

"Transaction Allowed.";

result.style.color="green";

}

}



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← → ↻ ⓘ File C:/Users/asd/OneDrive/Desktop/Ayush_katekhaye/C... ☆

☐ All B

Ayush Katekhaye

24070521161

ATM PIN Verification

....

Verify PIN

✓ **Strong PIN!**

Your PIN is NOT a palindrome.
Transaction Allowed.